

SCHOOL OF COMPUTER SCIENCE AND IT YBN UNIVERSITY RANCHI

JAVA PROGRAMMING ASSIGNMENT 16

1. String Concatenation and Length

- **Description:** Write a program that takes two strings from the user, concatenates them, and displays the concatenated string along with its length.
- **Example:** If the user inputs "Hello" and "World", the output should be "HelloWorld" with a length of 10.
- **Test Case:**
 - Input: "Java", "Programming"
 - Output: "JavaProgramming", Length: 15

2. Substring and Indexing

- **Description:** Develop a program that extracts a substring from a given string starting from a user-defined index *n* until the end of the string.
- **Example:** For the string "Programming" and index 4, the output should be "ramming".
- **Test Case:**
 - Input: "Welcome", 3
 - Output: "come"

3. String Comparison

- **Description:** Write a program to compare two strings and determine if they are identical. The comparison should be case-sensitive.
- **Example:** Comparing "apple" and "Apple" should indicate that the strings are not identical.
- **Test Case:**
 - Input: "Data", "Data"
 - Output: Strings are identical

4. Character Extraction

- **Description:** Create a program that prints every alternate character from a given string, starting with the first character.
- **Example:** For the string "abcdefg", the output should be "aceg".
- **Test Case:**
 - Input: "abcdefgh"
 - Output: "aceg"

5. Case Conversion

- **Description:** Write a program that converts all characters of a string from lowercase to uppercase and vice versa.
- **Example:** For the string "Hello World", the output should be "hELLO wORLD".
- **Test Case:**
 - Input: "JavaProgramming"
 - Output: "jAVApROGRAMMING"

6. Finding a Character in a String

- **Description:** Develop a program that checks if a specific character exists in a string and returns its first occurrence's index.
- **Example:** In the string "banana", if searching for "a", the output should be 1.
- **Test Case:**
 - Input: "mississippi", 's'
 - Output: 2

7. Replace Characters

- **Description:** Create a program to replace all occurrences of a given character in a string with another character specified by the user.
- **Example:** Replacing "a" with "o" in "banana" should give "bonono".
- **Test Case:**
 - Input: "Hello World", 'o', 'e'
 - Output: "Helle Werld"

8. Trimming Whitespaces

- **Description:** Write a program that trims leading and trailing spaces from a string entered by the user.
- **Example:** For the input string " Java Programming ", the output should be "Java Programming".
- **Test Case:**
 - Input: " Example String "
 - Output: "Example String"

9. Splitting Strings

- **Description:** Write a program that splits a string into an array of strings based on commas.
- **Example:** The string "apple,banana,cherry" should be split into ["apple", "banana", "cherry"].
- **Test Case:**
 - Input: "one,two,three,four"
 - Output: ["one", "two", "three", "four"]

10. Counting Specific Characters

- **Description:** Develop a program to count the number of times a specific character appears in a string.

- **Example:** The character "a" in "banana" should count as 3.
- **Test Case:**
 - Input: "parallel", 'l'
 - Output: 3